Nelson Kang

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Education

University of Waterloo

Bachelor of Computing and Financial Management (CFM)

Activities: Competitive Intramural Volleyball Captain, Poker Club Executive, Training for Ironman 70.3 Muskoka 2027
Awards: 1st of 18, Market Beat Portfolio Competition (2025)

Technical Skills

Languages: Python, C#, C, C++, SQL, TypeScript, JavaScript, HTML/CSS
Data & Infrastructure: AWS (S3, Athena, EventBridge, CodePipeline), Apache Iceberg, PyIceberg, DuckDB, Kafka, PostgreSQL, SQL Server, Redis, Docker, Terraform, Data Pipelines, CI/CD
Frameworks & Tools: FastAPI, Next.js, Celery, Modal, Supabase, Pandas, NumPy, Power BI, Postman, Git, Linux
ML / CV: Pytorch, YOLOv8, Roboflow, Machine Learning, OpenCV, FFmpeg

Experience

AI Engineer (Part-time) — Sep 2026 – Oct 2026

Skrypt Health — *Remote*

- Owned the **autonomous build harness** on **Amazon Bedrock**: cloud agent loops, peaking at **40 jobs a day**, took PRs to a ship verdict for human merge; jobs held no credentials and model calls were budget-capped.
- Shipped **lab results end to end** on a clinic's live athenahealth EHR: deterministic triage staff can audit, and a **write-back** of reviewed results to the chart, guarded by a pre-Save re-read.
- Built the **flagged-call loop**: AI calls clinics flag as wrong are labeled by root cause, answered with an approved reply, and auto-closed when the fix ships.
- Merged **99 PRs** across four repos in five weeks: fixed a **production outage** that had hidden every clinic support request, took a Windows test suite from 148 failures to 0, and cut CI cost by running drafts on a reduced suite.

Software Engineering Co-op — Jun 2026 – Aug 2026

Aquanow — *Toronto, ON*

- Fixed **5 critical production bugs** in a live regulatory pipeline (**Python, AWS S3/Athena, PyIceberg**) reporting to **Turkey's securities depository (MKK)**; restored reconciliation from **0 to all 107 currencies**.
- Extracted a shared **reconciliation library** with an **AI-agent workflow** that classifies and investigates breaks, reused it on a second domain with another co-op, and wrote the org-level **RFC and ADR** proposing it company-wide.
- Migrated company-wide month-end reconciliation onto **Apache Iceberg**, validating **3,800+** account identities.

Software Engineer Intern — Jun 2024 – Aug 2024

Ericsson Canada Inc. — *Ottawa, ON*

- Built **4 C#/SQL Server ETL pipelines** from Jira into reporting tables with **temporal** change history, and audited every team pipeline into the reference that became the team's standard **debugging resource**.

Projects

Skein — Decision Layer for Coding Agents — Oct 2026 – Present

Solo · TypeScript, MCP, Claude API, AWS — *skeinai.ca*

- Coding agents ask Skein before stopping for a person: it answers from the team's **recorded decisions** with a citation, or routes the question to the one person who owns it, and records the answer so the next ask settles.
- Built and deployed in two weeks by an **agent build loop** I designed (fresh-agent reviews, fixers, CI gates): **74 merged PRs**, plus a **replay eval** that scores each citation against the record as it stood when asked.

ClipFarm — Automated Volleyball Highlight Generator — Mar 2026 – Present

Founded and lead a team of 4 · Python, FastAPI, Celery, Redis, Modal, TypeScript, YOLOv8-pose — *clipfarm.ca*

- Turns a full match video into a filtered feed of **highlight clips**: a custom-trained ball detector bounds rallies by trajectory, cheer audio and rally-shape scoring keep the ones worth watching, and **YOLOv8-pose** labels each by action.
- Full stack: **FastAPI** endpoints with **Supabase JWKS** auth and **Row-Level Security**, **Celery + Redis** jobs, **Modal** GPU inference, and a **Next.js** clip grid whose correction UI feeds future training.

Dogwatch — Arctic Search Fleet Coordinator — Sep 2026

2nd Place, Dominion Dynamics WHITEOUT @ Hack the North 2026 ($1,000) · Python, MAVLink, ArduPilot

- Four machines (quadcopter, fixed-wing, two towers) search a **25 km strait** for an unlit vessel; a **probability field** counts every empty sweep as evidence. Built on my agent harness: **56 merged PRs** and **739 tests** in **29 hours**.